Pin Mapping Reference
Raspberry Pi Zero 2 W ↔ Adafruit 1651

2.8" TFT Touch Shield with Resistive Touch (TSC2007)

Custom PCB Design — DIY Kit

Point-to-point wiring between RPi header and 1651 shield header

Sources: Adafruit Learning System (learn.adafruit.com/adafruit-2-8-tft-touch-shield-v2), Raspberry Pi GPIO documentation (raspberrypi.com/documentation/computers/raspberry-pi.html#gpio)

RPi Zero 2 W GPIO Peripheral Reference

The following functions are fixed by the BCM2710A1 SoC and cannot be remapped in software. All GPIO pins operate at **3.3 V logic**. Applying 5 V to any GPIO pin will permanently damage the board.

Source: raspberrypi.com — GPIO documentation

PWM (pulse-width modulation)

- Software PWM available on all pins
- Hardware PWM available on **GPIO12, GPIO13, GPIO18, GPIO19**

SPI

- SPI0: MOSI (GPIO10); MISO (GPIO9); SCLK (GPIO11); CE0 (GPIO8); CE1 (GPIO7)
- SPI1: MOSI (GPIO20); MISO (GPIO19); SCLK (GPIO21); CE0 (GPIO18); CE1 (GPIO17); CE2 (GPIO16)

I2C

- Data: GPIO2 Clock: GPIO3
- EEPROM Data: GPIO0 EEPROM Clock: GPIO1 *(do not use for general I/O)*

Serial (UART)

- TX: GPIO14 (physical pin 8) RX: GPIO15 (physical pin 10)
- *These are reserved for the Linux serial console by default. Avoid using them for general I/O in a DIY kit.*

Power and Ground Pin Notes

5V pins — Pin 2 and Pin 4 (both even side): Both pins are electrically identical — connected directly to the incoming micro-USB supply rail with no intervening regulation or current limiting beyond the board's polyfuse. The duplication is for PCB routing convenience only. Either may be used; for a DIY kit, Pin 2 is conventional. Current capacity is limited only by your power supply and the board's polyfuse (typically rated around 2 A for the Zero 2 W).

3.3V pins — Pin 1 (odd) and Pin 17 (odd): Both are outputs of the same onboard 3.3 V regulator. They are electrically identical. The **total** current available from the 3.3 V rail (shared with all GPIO output drivers and any attached peripherals) is approximately **50 mA**. The 1651 shield has its own onboard 3.3 V regulator fed from the 5 V rail, so it does *not* draw from the RPi's 3.3 V supply. Do not power the 1651 from the RPi 3.3 V pin.

Ground pins: Physical pins 6, 9, 14, 20, 25, 30, 34, and 39 are all ground. They are all identical. Use the ground pin physically closest to the signal pins being routed for best noise performance. For a two-connector PCB layout (odd-side and even-side headers), use Pin 9 or Pin 25 for the

odd-side (1651) ground return, and Pin 6, 14, or 20 for the even-side ground return.

RPi Zero 2 W — Complete Pin Allocation Table

Legend: Odd-side pin Even-side pin Power / Ground

Column headers: **Phys** = physical header pin number; **BCM** = Broadcom GPIO number used in Python code; **Dir** = I=input, O=output, PWM, PWR; **Assignment** = function in this design.

Phys	BCM	Dir	Assignment	Notes
Odd-side pins — 1651 Display and Touch (point-to-point wiring to shield)				
1	—	PWR	3.3 V	*Not used for 1651. Available for other 3.3 V loads if total rail current stays under 50 mA.*
3	GPIO2	I/O	I2C SDA → TSC2007	**FIXED** by SoC. I2C bus 1 data. Enable with `dtparam=i2c_arm=on` in `/boot/config.txt`. The 1651 shield has onboard 4.7 kΩ pull-ups on SDA/SCL — do not add additional pull-ups on the PCB. See protective resistor note (Section 4).
5	GPIO3	I/O	I2C SCL → TSC2007	**FIXED** by SoC. I2C bus 1 clock. Same pull-up note as SDA above.
7	GPIO4	I	TSIRQ (touch interrupt)	*soft* assignment. Active-low interrupt from TSC2007. Enable internal pull-up in software: `GPIO.setup(4, GPIO.IN, pull_up_down=GPIO.PUD_UP)`. Add 33 Ω series resistor on PCB for ESD protection.
9	—	GND	Ground (1651 return)	Use for 1651 shield ground return. Close to SPI pins.
11	GPIO1	O	SD Card CS	*soft* assignment. Active-low chip select for microSD on 1651. Add 33 Ω series resistor on PCB.
17	—	PWR	3.3 V	*Not used for 1651. See power notes above.*
19	GPIO1(	O	SPI0 MOSI → ILI9341	**FIXED** by SoC (SPI0 MOSI). High-speed data to display. Add 33 Ω series resistor on PCB.
21	GPIO9	I	SPI0 MISO ← ILI9341	**FIXED** by SoC (SPI0 MISO). Read-back from display/SD. Add 33 Ω series resistor on PCB.
23	GPIO1	O	SPI0 SCLK → ILI9341	**FIXED** by SoC (SPI0 clock). Add 33 Ω series resistor on PCB.
25	—	GND	Ground	Additional ground; use if needed for SPI return path.
27	GPIO0	—	*EEPROM — DO NOT USE*	Reserved for HAT ID EEPROM. Leave unconnected.
29	GPIO5	—	*Spare*	Unallocated. Reserve for future use.

Phys	BCM	Dir	Assignment	Notes
33	GPIO1	PWM	Backlight PWM → 1651 LITE	*soft* assignment. **Hardware PWM channel 1** (BCM2710 PWM1). Connect to 1651 LITE pin (solder the LITE/#3 jumper on shield). Add 33 Ω series resistor on PCB. Software: use `RPi.GPIO` PWM or `pigpio` for precise frequency.
35	GPIO1	O	TFT DC (Data/Command)	*soft* assignment. Data/Command select for ILI9341. Note: GPIO19 is also SPI1 MISO — not a conflict since SPI1 is unused. Add 33 Ω series resistor on PCB.
37	GPIO2	—	*Spare*	Unallocated. Reserve for future use.
39	—	GND	Ground	Spare ground, available for PCB routing.
Even-side pins — General I/O (LEDs, Inputs, 1-Wire, MOSFET control)				
2	—	PWR	5 V → 1651 VIN	**Primary 5 V supply for the 1651 shield.** The shield's onboard regulator converts this to 3.3 V for the ILI9341 and TSC2007. Do not use the RPi 3.3 V rail to power the 1651.
4	—	PWR	5 V spare	Electrically identical to Pin 2. Available for other 5 V loads on the PCB (e.g., MOSFET gate driver supply if needed).
6	—	GND	Ground (even-side return)	Primary ground return for even-side I/O.
8	GPIO1	—	*UART TX — avoid*	Linux serial console TX by default. Do not use for general I/O in a kit. Leave unconnected at header.
10	GPIO1	—	*UART RX — avoid*	Linux serial console RX by default. Leave unconnected at header.
12	GPIO1	O	Indicator LED 3	*soft* assignment. Hardware PWM capable (PWM0) if dimming desired. Also SPI1 CE0 — not a conflict since SPI1 is unused. Add 330 Ω current-limit resistor in series with LED on PCB.
14	—	GND	Ground	Additional even-side ground.
16	GPIO2	O	Indicator LED 2	*soft* assignment. General GPIO output. Add 330 Ω current-limit resistor in series with LED on PCB.
18	GPIO2	I	Digital Input 1 (Button A)	*soft* assignment. Enable internal pull-up in software. Add 33 Ω series ESD resistor on PCB. External button connects GPIO to GND.
20	—	GND	Ground	Additional even-side ground.
22	GPIO2	I	Digital Input 2 (Button B)	*soft* assignment. Enable internal pull-up in software. Add 33 Ω series ESD resistor on PCB.

Phys	BCM	Dir	Assignment	Notes
24	GPIO8	O	TFT CS → ILI9341 (SPI0 CE0)	**FIXED** by SoC (SPI0 CE0). **Only even-pin forced exception for 1651.** Hardware-managed chip select. Add 33 Ω series resistor on PCB.
26	GPIO7	I/O	1-Wire bus	*soft* assignment (non-default pin). **Requires:** `dtoverlay=w1-gpio,gpiopin=7` in `/boot/config.txt`. **Requires 4.7 kΩ pull-up to 3.3 V on PCB** (this is a PCB passive component, not a GPIO internal pull-up — the 1-Wire protocol requires a strong external pull-up). GPIO7 is also SPI0 CE1 — not a conflict since CE1 is unused in this design. Note: standard default 1-Wire pin is GPIO4 (physical pin 7, odd side). Moving to GPIO7 requires the dtoverlay shown above.
28	GPIO1	—	*EEPROM — DO NOT USE*	Reserved for HAT ID EEPROM. Leave unconnected.
30	—	GND	Ground	Additional even-side ground.
32	GPIO1	O	Indicator LED 1	*soft* assignment. Hardware PWM capable (PWM0 channel A) if dimming desired. Add 330 Ω current-limit resistor in series with LED on PCB.
34	—	GND	Ground	Additional even-side ground.
36	GPIO1	O	Control LED 1 (MOSFET gate)	*soft* assignment. GPIO output to MOSFET gate driver. Note: GPIO16 is SPI1 CE2 — not a conflict since SPI1 unused. Add 33 Ω gate resistor to MOSFET on PCB.
38	GPIO2	O	Control LED 2 (MOSFET gate)	*soft* assignment. GPIO output to MOSFET gate driver. Add 33 Ω gate resistor on PCB.
39	—	GND	Ground	See odd-side row above.
40	GPIO2	O	Control LED 3 (MOSFET gate)	*soft* assignment. GPIO output to MOSFET gate driver. Add 33 Ω gate resistor on PCB.

Remaining allocations not yet assigned (4 inputs, 2 MOSFET channels): The current allocation places 3 of 5 MOSFET/LED control outputs (GPIO16, 20, 21) and 2 of 4 inputs (GPIO24, 25) and 3 indicator LEDs (GPIO12, 18, 23) on the even side. Two more MOSFET outputs and two more inputs still need assignment. Suggested even-side candidates: GPIO22 (pin 15 — odd, spare after LED move), or confirm the final even-side spare pins with the designer before routing. *Finalise before PCB layout.*

Adafruit 1651 Shield Pin Mapping Table

Point-to-point wiring from each shield pin to the RPi header. Column 1 is the Arduino/Metro silkscreen label on the 1651 shield. Passive components (series resistors) sit on the PCB between the RPi header and the wire going to the 1651 shield pin.

1651 Pin (Arduino label)	RPi Odd (physical/B	RPi Even (physical/B	Description	Notes
5V / VIN	—	Pin 2	Board power in	5 V from RPi micro-USB rail. Feeds onboard 3.3 V regulator. Do *not* use RPi 3.3 V. No series resistor — power line. Use adequate trace width on PCB.
GND	Pin 9 or 25	Pin 6, 14, or 20	Ground	Common ground. Use the ground pin nearest the signal traces being routed.
ICSP MOSI	Pin 19 / GPIO10	—	SPI0 MOSI → ILI9341	**FIXED**. 33 Ω series resistor on PCB (RPi side of resistor connects to pin 19; shield side connects to ICSP MOSI pin).
ICSP MISO	Pin 21 / GPIO9	—	SPI0 MISO ← ILI9341	**FIXED**. 33 Ω series resistor on PCB.
ICSP SCK	Pin 23 / GPIO11	—	SPI0 clock	**FIXED**. 33 Ω series resistor on PCB.
D10 — TFT CS	—	Pin 24 / GPIO8	Display chip select (CE0)	**FIXED** (SPI0 CE0). Only even-side pin forced for 1651. 33 Ω series resistor on PCB. Active-low.
D9 — TFT DC	Pin 35 / GPIO19	—	Data/Command select	*soft*. 33 Ω series resistor on PCB.
D4 — SD CS	Pin 11 / GPIO17	—	microSD chip select	*soft*. Active-low. 33 Ω series resistor on PCB. Only wire if microSD is used; otherwise leave open.
SDA	Pin 3 / GPIO2	—	I2C SDA → TSC2007	**FIXED**. The 1651 shield has onboard 4.7 kΩ pull-up. Do *not* add a second pull-up on PCB. 33 Ω series resistor on PCB for ESD.
SCL	Pin 5 / GPIO3	—	I2C clock → TSC2007	**FIXED**. Same pull-up and ESD notes as SDA.
D2 — TSIRQ	Pin 7 / GPIO4	—	Touch interrupt (active-low)	*soft*. Solder the TSIRQ/#2 jumper on 1651 to enable. Enable RPi internal pull-up in software. 33 Ω series resistor on PCB.
D3 — LITE	Pin 33 / GPIO13	—	Backlight PWM	*soft*. Solder the LITE/#3 jumper on 1651 to enable. Hardware PWM channel 1. 33 Ω series resistor on PCB.
VddIO jumper	**IMPORTANT:** The 1651 VddIO jumper must be set to **3V** (solder left pad to centre pad) because the RPi Zero 2 W is a 3.3 V logic device. This selects the logic reference voltage for the 74LVX245 level shifters on the shield. Incorrect setting risks damaged GPIO pins.			

Passive Component Recommendations for DIY Kit

Protective Series Resistors on 1651 Signal Lines

All signal lines between the RPi GPIO header and the 1651 shield header should have a **33 Ω series resistor** placed on the PCB, on the RPi side of the wire run (i.e., close to the RPi header).

Rationale: The RPi's GPIO pins have no internal current limiting or ESD protection diodes rated for external misconnection. A 33 Ω resistor limits the peak fault current if a signal line is accidentally shorted to 5 V or GND during kit assembly or use, and damps transmission-line ringing on the SPI clock and data lines at high speed (40 MHz). At 40 MHz SPI, the RC time constant with typical PCB trace capacitance (∼5 pF) gives a corner frequency well above 100 MHz, so the resistor has negligible effect on signal integrity at this speed.

Value: 33 Ω is the standard recommended value for RPi GPIO protection. Acceptable range: 22–47 Ω. **Package:** 0402 or 0603 SMD preferred for a compact PCB. Place as close as possible to the RPi GPIO header pin.

Specific lines and resistor placement:

- **SPI lines (MOSI, MISO, SCLK, TFT CS, TFT DC, SD CS):** 33 Ω each. Place within 5 mm of RPi header.
- **I2C lines (SDA, SCL):** 33 Ω each. Note: the 1651 shield already has 4.7 kΩ pull-ups on these lines. Do *not* add PCB pull-ups — the shield pull-ups are sufficient.
- **TSIRQ:** 33 Ω series. RPi internal pull-up enabled in software.
- **LITE (backlight PWM):** 33 Ω series.
- **5 V power line:** No series resistor. Use adequate trace width (≥0.5 mm for up to 500 mA).
- **GND:** No series resistor.

1-Wire Bus Pull-up

The 1-Wire bus on GPIO7 (physical pin 26) **requires a 4.7 kΩ pull-up resistor to 3.3 V** on the PCB. This is a hard requirement of the 1-Wire protocol — the internal RPi pull-up (50 kΩ) is far too weak and will cause communication failures. Use a 4.7 kΩ resistor to the 3.3 V rail (RPi pin 1 or pin 17). Package: 0402 or 0603 SMD.

LED Current-Limit Resistors

All LED outputs (indicator LEDs on GPIO12, GPIO18, GPIO23; control LEDs on GPIO16, GPIO20, GPIO21) must have current-limiting resistors in series with the LED on the PCB.

- RPi GPIO output voltage: 3.3 V
- Maximum current per GPIO pin: 16 mA (recommended: 8–12 mA for longevity)
- Typical LED forward voltage: 2.0 V (red/yellow), 3.0 V (blue/green/white)
- **Recommended resistor for red/yellow LED:** $(3.3 - 2.0)\,\text{V}/10\,\text{mA} = 130\,\Omega$, use **150 Ω**
- **Recommended resistor for blue/green/white LED:** $(3.3 - 3.0)\,\text{V}/10\,\text{mA} = 30\,\Omega$, use **33 Ω** (or higher if the LED is very bright)
- Package: 0402 or 0603 SMD.

MOSFET Gate Resistors

For MOSFET gate drive lines (GPIO16, GPIO20, GPIO21, and any remaining MOSFET control pins), place a **33–100 Ω gate resistor** in series between the GPIO output and the MOSFET gate. This limits the gate charge current spike at switch-on, which would otherwise present a momentary short-circuit load to the GPIO pin. If MOSFETs switch inductive loads, add a flyback diode across the load.

Decoupling Capacitors

On the RPi header power pins: Place a **100 nF ceramic capacitor** (0402 or 0603, X5R or X7R) between the 5 V supply pin (pin 2) and the nearest GND pin (pin 6) on the PCB, as close as possible to the connector. This decouples switching transients on the supply caused by the display's SPI bursts.

On the 1651 shield header power pin: Place a **100 nF ceramic** plus optionally a **10 µF electrolytic** (or tantalum) in parallel, between VIN and GND at the shield power header pins. The larger capacitor absorbs the inrush current when the display backlight and SPI bus first activate.

On the 3.3 V rail (for 1-Wire and I2C pull-ups): Place a **100 nF ceramic** between 3.3 V (pin 1 or 17) and GND close to the pull-up resistors.

No capacitors are needed in the GPIO signal paths. The 33 Ω series resistors already damp any ringing. Adding capacitance to SPI lines would slow the edges at high speed.

Software Configuration Notes

- **SPI:** Enable with `dtparam=spi=on` in `/boot/config.txt`
- **I2C:** Enable with `dtparam=i2c_arm=on` in `/boot/config.txt`
- **1-Wire:** Enable with `dtoverlay=w1-gpio,gpiopin=7` in `/boot/config.txt` (non-default pin; standard default is GPIO4)
- **Backlight PWM:** GPIO13 is hardware PWM channel 1 (PWM1). Use `pigpio` library for precise frequency and duty-cycle control, or `RPi.GPIO` for simple on/off and software PWM.
- **TSIRQ:** Configure GPIO4 as input with pull-up. TSC2007 default I2C address: `0x48`.
- **1651 VddIO jumper:** Must be set to 3V before powering on.

Summary: Unallocated and Reserved Pins

Physical	BCM	Status
Pin 1	—	3.3 V — available for low-current peripherals
Pin 17	—	3.3 V — available for low-current peripherals
Pin 13	GPIO27	*Spare odd-side GPIO — reserve*
Pin 15	GPIO22	*Spare odd-side GPIO — reserve*
Pin 27	GPIO0	EEPROM — do not use
Pin 28	GPIO1	EEPROM — do not use
Pin 8	GPIO14	UART TX — avoid in kit
Pin 10	GPIO15	UART RX — avoid in kit

Outstanding allocation items — finalise before PCB layout:

- 2 remaining MOSFET control outputs (of 5 total; 3 assigned: GPIO16, 20, 21)
- 2 remaining digital inputs (of 4 total; 2 assigned: GPIO24, 25)
- Candidates: even-side GPIO22 (pin 15 odd — reconsidered if moved), or confirm further even-side spares with designer
- Confirm whether all 5 control LEDs vs MOSFET gate signals share the same pins, or whether LED indicators and MOSFET gates are on separate GPIO lines